Recommendation Scores

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Recall and choose your starting point

> Record your starting explanation for Recommendation Scores before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Nearest Neighbor Reasoning

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Recommendation Scores trace and record the first step you can explain without help.

Use W2: Check a boundary for **Weight**ed **score**s and **constraint**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Recommendation Scores explanation against the supplied facts before discussing it.

Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Choose independently

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Recommendation Scores target to a trace and a justified conclusion.

> **Score**=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Recommendation Scores case and show the deciding evidence.

Score=2 times genre match + length match. A matches both, B genre only, C length only.

Score and **rank**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute **score**s and justify the **rank**ing.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 23 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Recommendation Scores adds to Transparent Book Recommender and keep the evidence.

Publish the features, **weight**s, raw **score**s, and reason for the top result.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Score=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Calculate a score

Calculate a score. Apply the supplied Recommendation Scores facts and keep your reasoning.

An eligible activity has fun 2 and learning 4. Use 2 * fun + learning.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Check a boundary

Check a boundary. Apply the supplied Recommendation Scores facts and keep your reasoning.

Which durations 19, 20, 21 qualify under at most 20?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair selection

Repair selection. Apply the supplied Recommendation Scores facts and keep your reasoning.

An ineligible item **score**s 15 and an eligible item **score**s 9. Which should be offered under the stated policy?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Choose independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Choose independently. Apply the supplied Recommendation Scores facts and keep your reasoning.

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

> **My reasoning and evidence**

W5 Test sensitivity

> Test sensitivity. Apply the supplied Recommendation Scores facts and keep your reasoning.

For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare **weight**s (2,1) with (1,2).

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Recommendation Scores

> A fictional book-activity chooser has a 14-minute limit. Explain the labelled step before checking the changed condition.

A fictional book-activity chooser has a 14-minute limit.

Score=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

AI ACADEMY / AI-2/23

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/23

Recommendation Scores

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Eligibility first

A and C fit time; B takes 15 and is excluded.

2. Scores

A:2 x 3+2=8. C:2 x 2+5=9.

3. Choose

C wins at score 9 among eligible activities.

4. Explain exclusion

B's score 11 cannot override its failed time requirement.

CHANGE ONE THING

Change interest weight from 2 to 3, keeping time limit 14.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why does B not become the winner when its **score** is largest?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-activity chooser has a 14-minute limit.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B's **score** 11 cannot override its failed time requirement.

> 1. Choose

2. Explain exclusion

3. Eligibility first

4. **Score**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Explain exclusion B's **score** 11 cannot override its failed time requirement.

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change interest **weight** from 2 to 3, keeping time limit 14. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A **score**s 11; C also **score**s 11; tie-by-ID chooses A. B remains too long; changing preferences changes only **rank**ing of A and C.

Repair a missing building block

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and **rank training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

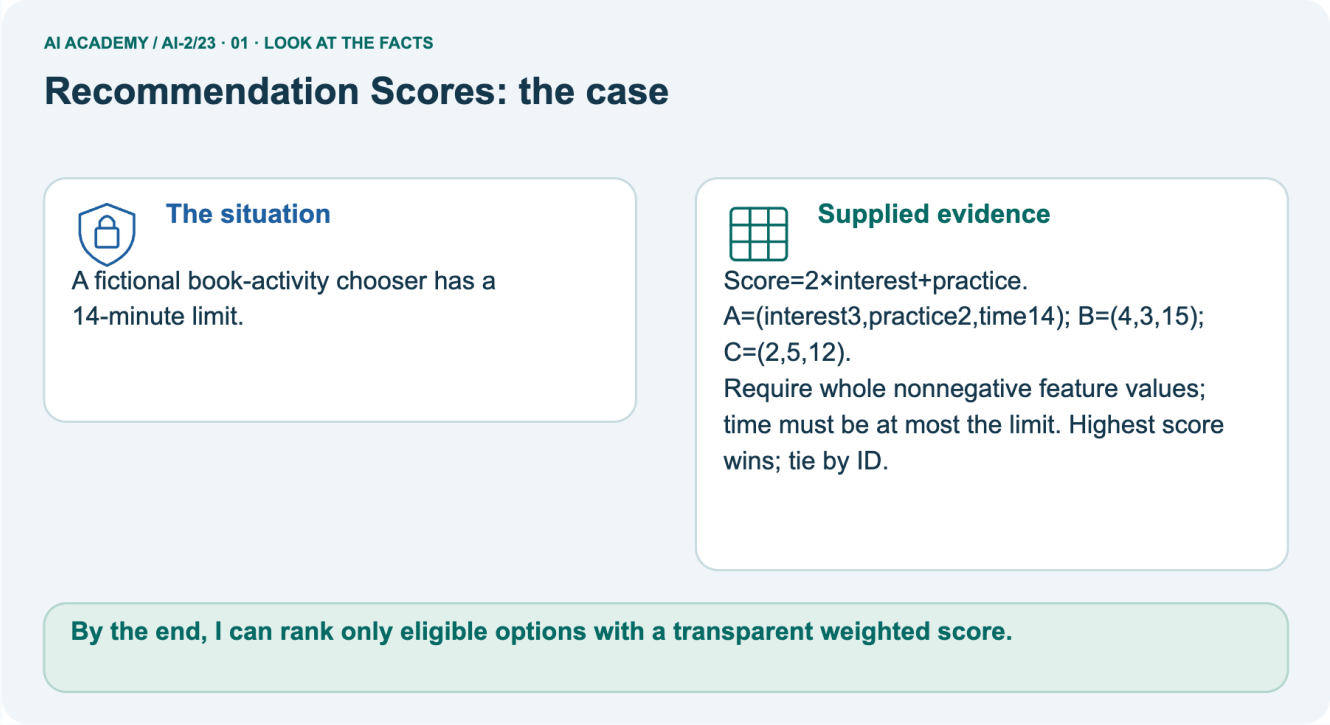
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Recommendation Scores picture from its labels, then record a relevant limit.

01. Recommendation Scores: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

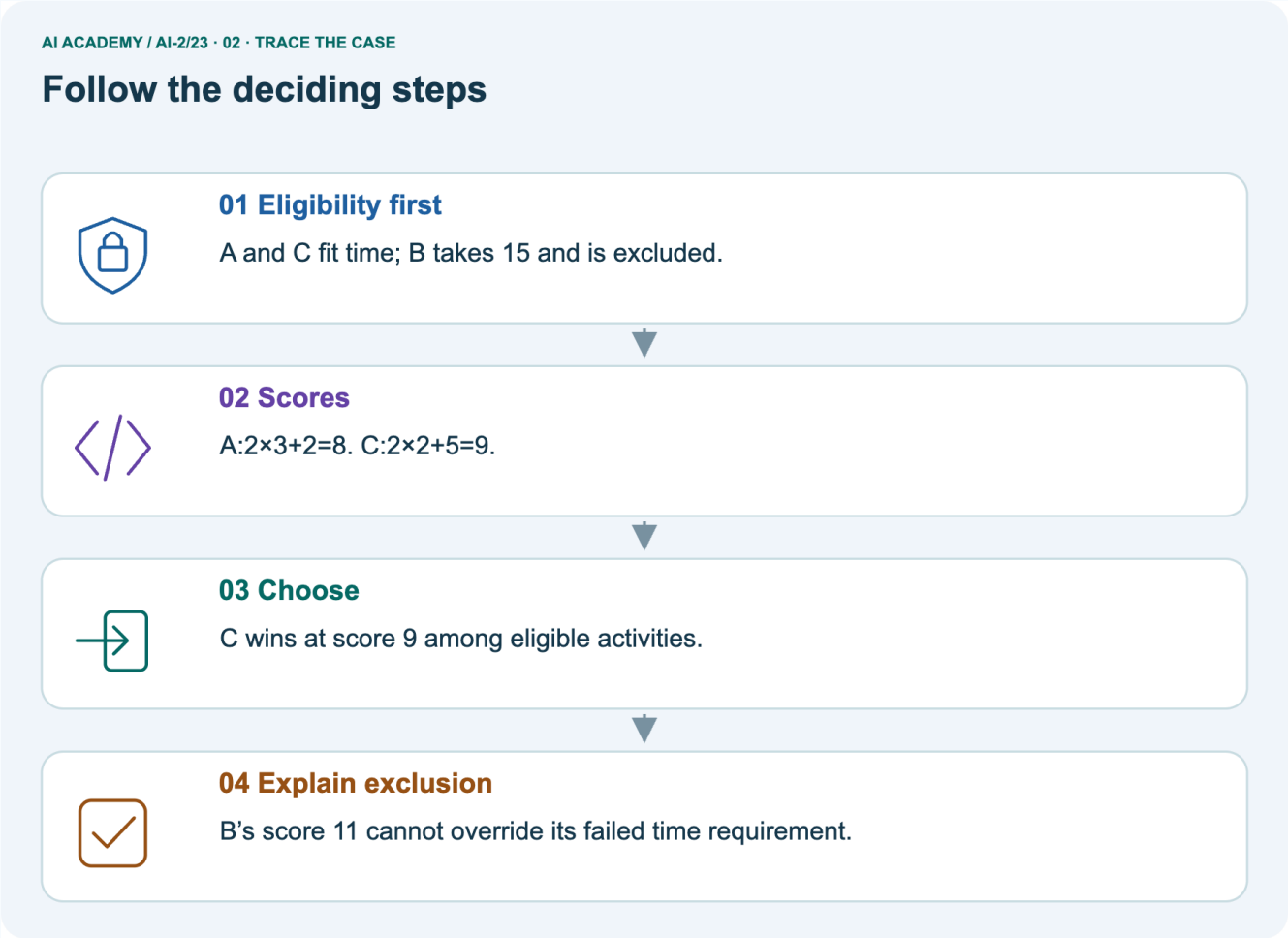
A fictional book-activity chooser has a 14-minute limit. Supplied facts: **Score**=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility first: A and C fit time; B takes 15 and is excluded. → **Score**s: A:2×3+2=8.

C:2×2+5=9. → Choose: C wins at **score** 9 among eligible activities. → Explain exclusion: B's **score** 11 cannot override its failed time requirement.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Time filtering and scores



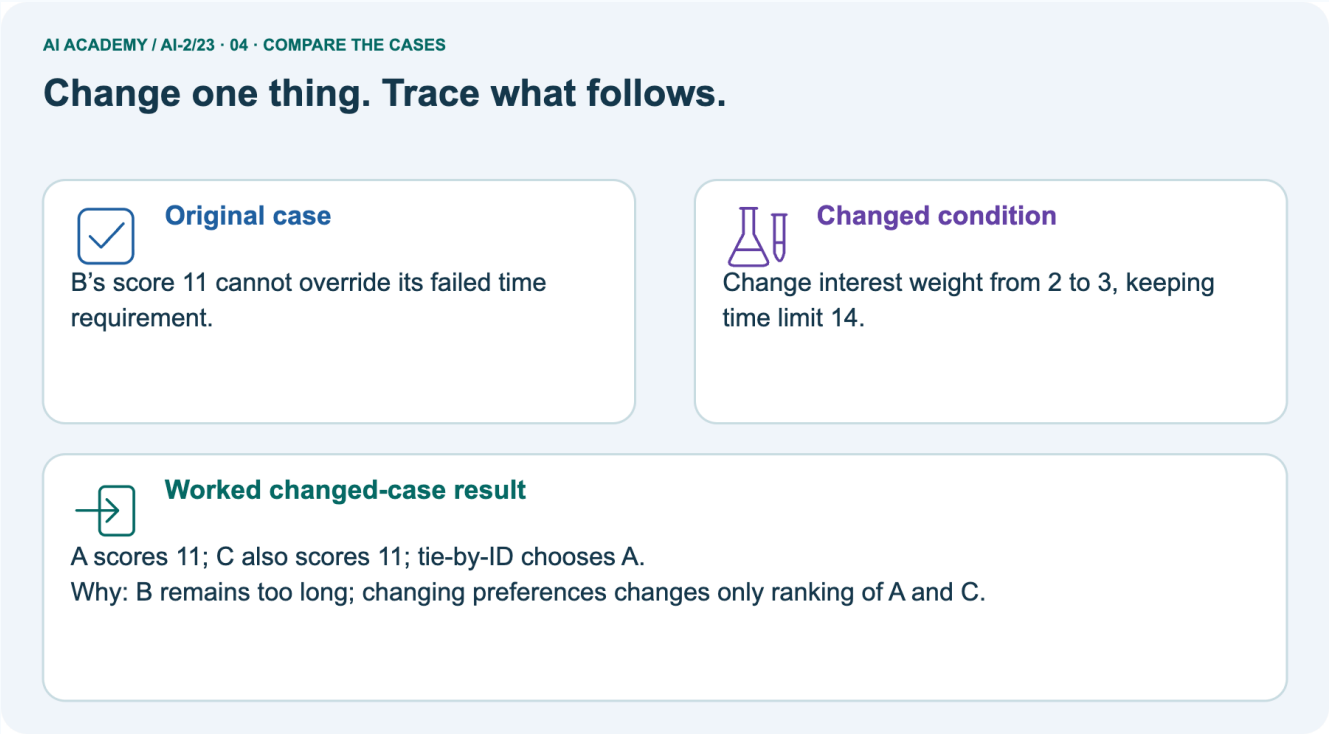
Follow the labelled picture with the matching explanation. Open the vector diagram

Time filtering and **score**s. A: Minutes = 14, **Score** = 8, Status = eligible. B: Minutes = 15, **Score** = 11, Status = excluded. C: Minutes = 12, **Score** = 9, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

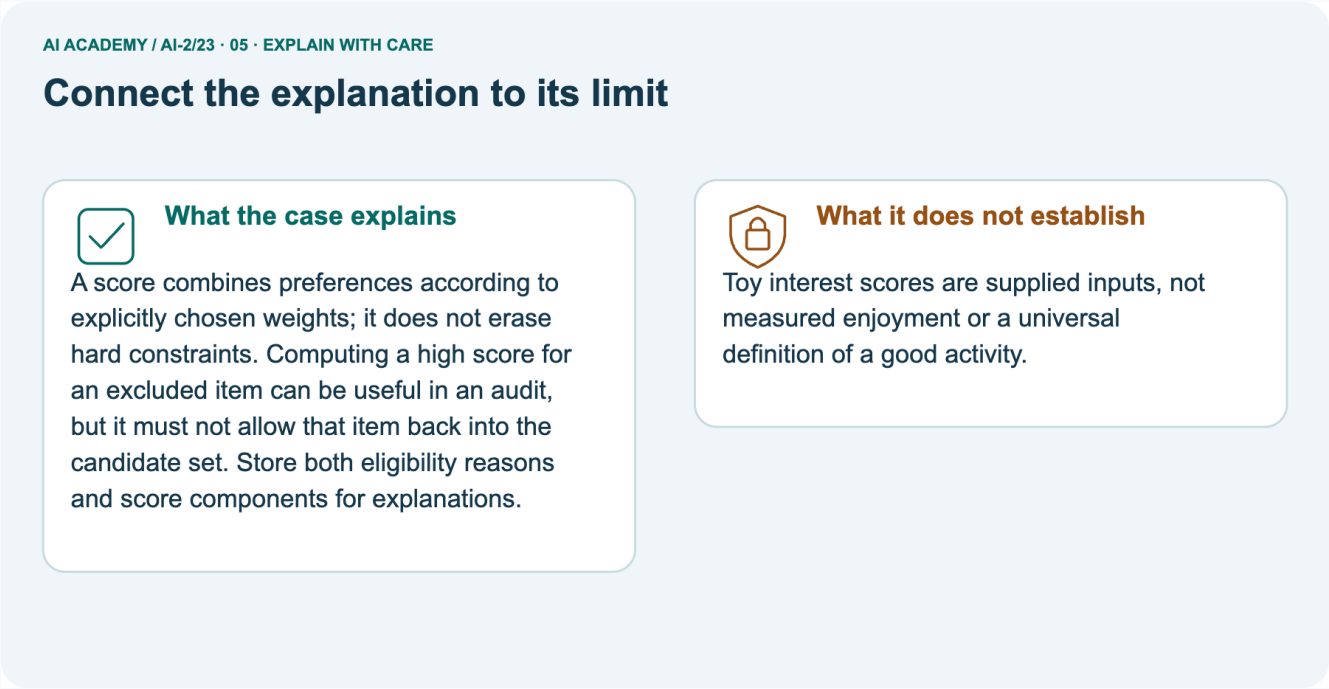
Original case: B's **score** 11 cannot override its failed time requirement. Changed condition: Change interest **weight** from 2 to 3, keeping time limit 14. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

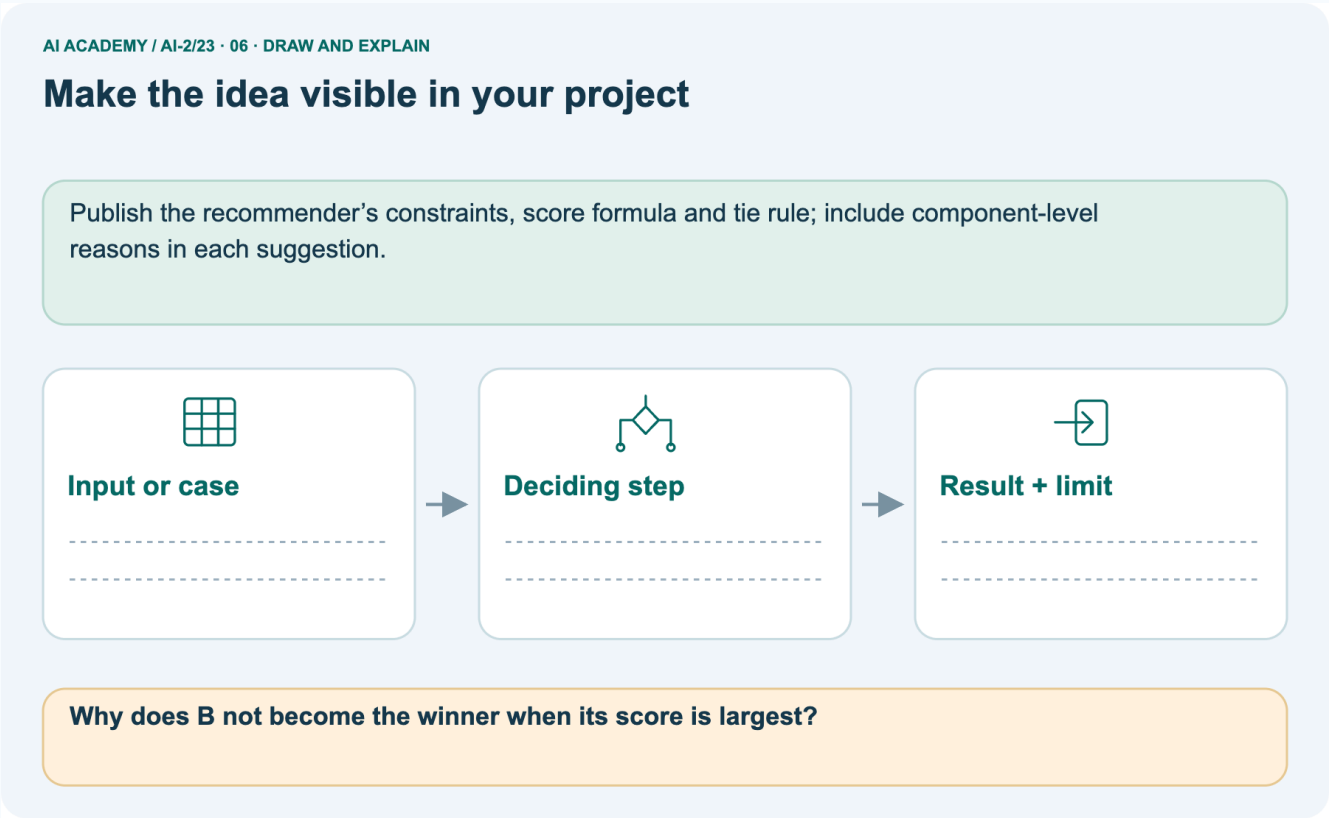
Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Limit: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion. Draw your **input**, deciding step, result and limit.

Supported question: Why does B not become the winner when its **score** is largest?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.